SCIENCE OLYMPIAD FOUNDATION

Class 8

IEO

Set B

Year 2018

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FOR QUESTIONS 1 TO 24, CHOOSE THE CORRECT WORD/PHRASE TO COMPLETE EACH SENTENCE.

Example

Ice floats if you _____ it in water.

A. drops
B. drop
C. dropped
D. can drop

1. _____ is very slow unfortunately, but it should speed up really soon.
   A. Developing
   B. Development
   C. Developed
   D. Developmental

2. We really like _____ of the new classes this year.
   A. these
   B. one another
   C. several
   D. any

3. My friends _____ football until late last night.
   A. play
   B. have played
   C. are playing
   D. were playing

4. I _____ maths all weekend until I realized I had a history test on Monday.
   A. had been studying
   B. have studied
   C. learned
   D. was learning

5. He can speak Dutch _____, which is quite rare for English people.
   A. fluently
   B. outside
   C. often
   D. quite

6. The _____ finalists walked onto the field proudly.
   A. glorify
   B. glorious
   C. glorification
   D. glory

7. The car keys are _____ top of the fridge _____ the papers.
   A. at, in
   B. in, next to
   C. by, at
   D. on, under

8. The new captain walked through the doors as _____ he was a king.
   A. though
   B. so
   C. unless
   D. however

9. Things will never be the same after the changes you made, _____?
   A. do they
   B. will they
   C. are they
   D. can they

10. Brian told me that he had _____ football all day yesterday until it rained.
    A. been played
    B. played
    C. been playing
    D. was playing

11. I've had this carpet that I bought from Mumbai for about 10 years, _____ I?
    A. don't
    B. isn't
    C. won't
    D. haven't

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12. _____ that I get to the finals, I doubt I will be champion.
    A. Assuming
    B. Attacking
    C. Thinking
    D. Removing

13. Those vegetarian dishes, especially _____ you cooked yesterday, were very tasty.
    A. which
    B. the one
    C. how
    D. so much

14. _____ Chinese people often have dinner late at night.
    A. A
    B. An
    C. The
    D. No article

15. Deciding _____ was a very bad idea; it's too far.
    A. walking
    B. walk
    C. to walk
    D. to have walked

16. Don't worry, it will work; there's method to his _____.
    A. madly
    B. mad
    C. maddeningly
    D. madness

17. I need to get home, so I can get these vegetables in the fridge before they go _____.
    A. up
    B. off
    C. out
    D. on

18. Somebody needs to _____ and claim this prize or it will go to waste.
    A. run into
    B. sit down
    C. step up
    D. jump over

19. The flowers _____ so much in the past 3 days.
    A. have grown
    B. grow
    C. are growing
    D. grown

20. I _____ never been any good at football until I learnt this simple trick.
    A. had
    B. have
    C. having
    D. have had

21. The ceiling is _____ quite a bit, so I think we should get an engineer in.
    A. cramping
    B. reviewing
    C. enlisting
    D. subsiding

22. I live in the same _____ area as all my friends, we all go to the same school.
    A. intake
    B. catchment
    C. education
    D. living

23. I _____ for a second and the clouds came over.
    A. ran
    B. dozed
    C. encountered
    D. relived

24. This is a very _____ club; absolutely anyone can be a member.
    A. edible
    B. elite
    C. catholic
    D. destructive

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FOR QUESTIONS 25 AND 26, CHOOSE THE CORRECT SYNONYM OF THE GIVEN WORD.

25. Empower
    A. Authorise
    B. Explain
    C. Dissolve
    D. Trash

26. Mutate
    A. Emulate
    B. Change
    C. Trade
    D. Subsume

FOR QUESTIONS 27 AND 28, CHOOSE THE CORRECT ANTONYM OF THE GIVEN WORD.

27. Purify
    A. Pollute
    B. Develop
    C. Remove
    D. Pre-order

28. Scrutiny
    A. Remove
    B. Crush
    C. Glance
    D. Dunk

FOR QUESTIONS 29 AND 30, CHOOSE THE OPTION WITH CORRECT SPELLING.

29. What is the spelling of the word that means 'giving and kind'?
    A. Megnanimous
    B. Magananimous
    C. Maganimious
    D. Magnanimous

30. What is the spelling of the word that means 'having healing powers'?
    A. Therapeutic
    B. Thereputeic
    C. Theriputic
    D. Theropuetic

READING

FOR QUESTIONS 31 TO 35, READ THE PASSAGE AND ANSWER THE QUESTIONS THAT FOLLOW.

With climate change and overfishing wrecking havoc on our oceans, it is becoming increasingly important for researchers to closely monitor our marine life. However, observing sea creatures up close is almost impossible since human presence scares the animals. Now, thanks to The Soft Robotic Fish, aka SoFi, researchers may not only be able to keep a close eye on the elusive creatures, but also uncover undersea secrets that have been eluding us for centuries.

Built by MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL), the snow white remote-controlled robot closely emulates real fish, complete with a flexible tail that flicks from side to side and two "fins". Though not the first autonomous underwater vehicle (AUV) created to monitor the oceans, SoFi resolves many of the issues that have hindered the usefulness of previous robots.

AUV's traditionally have had to be tied to a boat because radio frequency communications do not work well underwater. To overcome the problem they used sound waves. This technology can travel greater distances allowing drivers, using a modified, waterproofed Super Nintendo controller, to pilot SoFi from up to 50 feet away. To ensure SoFi can freely move around the ocean and maneuver like a real fish, a hydraulic pump seamlessly transfers water from one balloon-like structure to the other through its soft rubber tail, while a controller fitted with a smartphone battery powers the robot

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through the ocean. CSAIL Ph.D. candidate Robert Katzschmann, lead author of the study says, "To our knowledge, this is the first robotic fish that can swim freely for extended periods of time."

Also limiting traditional robot usefulness is the risk of crashing. With an outside made of rubber and flexible plastic that keeps its in-built electronics dry, SoFi poses no such problem. "Collision avoidance often leads to inefficient motion, since the robot has to settle for a collision-free trajectory," says Rus. "In contrast, a soft robot is not just more likely to survive a collision but could use it as information to inform a more efficient motion plan next time around."

While SoFi currently only records video, future versions will include sensors, such as thermometers. The researchers also hope to make it more **autonomous** and see a day when swarms of the soft robots, powered by solar cells, will follow fish around, allowing scientists to gain insights into schooling dynamics and monitor their population.

31. Choose the best title or heading for the passage.
    A. Robot Fish Monitor the Sea
    B. Nature and Robots Work Together
    C. A Robot to Form the Seabed
    D. New Robot Eats Fish for Energy

32. What do people call the robot?
    A. Robert
    B. AUV
    C. CSAIL
    D. SoFi

33. How does the robot move around underwater?
    A. Using a rope
    B. Using a pump
    C. Using a propeller
    D. Using another boat

34. What problem has the soft robot solved?
    A. Disappearing
    B. Sinking
    C. Crashing
    D. Diving

35. What does the word 'autonomous' mean in the last paragraph?
    A. With help
    B. Independent
    C. By wire
    D. Badly

FOR QUESTIONS 36 TO 40, READ THE PASSAGE AND ANSWER THE QUESTIONS THAT FOLLOW.

While an entire month free of homework or tests may sound too good to be true, that is precisely what the students at Sturenskolan School in Boden, Sweden were treated to in April. What's more, if the results of the experiment prove encouraging, starting 2019, the middle schoolers will never have to worry about after-school work or tests, aside from the national examinations, which are mandatory for students across the country.

The school's principal, Petronella, who came up with the idea, told Swedish national broadcaster, "We are constantly receiving new reports that our children feel bad. And that's because they have a very high-stress level all the time. This is our way of trying to reduce that stress."

The educator says that while banning homework sounds radical, it makes sense given the changes in curriculum and approach to education. While in the past students were primarily subjected to memorising and producing facts, learning today is all about reflection and analysis. As a result, it is often harder for kids, especially those with little support at home, to understand the material by themselves. Hence, it is only fair to grade them based on their class work.

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Sweden is not the only country that believes after-school work does not help students learn. In March 2018, an elementary school in Montreal, Canada, also decided to do away with homework. They hope parents will spend the extra time reading to their kids.

The debate over how much, if any, homework kids should be given is not new. In their 2006 book 'The Case Against Homework', based on American research argues that while there is no proof of the benefits of homework, there is plenty of evidence of its adverse effects. A 2014 Stanford University study conducted on 4,000 students revealed that the daily stress of after-school work not only leads to decreased interaction with friends and family, but also results in sleeping problems and physical ones, such as headaches.

While the long-term impact of the recent decisions made by the two educators remains to be seen, the experiment has already proved **efficacious** in Finland. The Nordic country, which is at the forefront of education reform, has done away with homework, grades, and even teaching by subjects for many years. Yet, Finnish students always rank high in the PISA (Programme for International Student Assessment) tests especially in maths and science!

36. Choose the best title or heading for the passage.
    A. Just Like Canada and Finland
    B. Free School Lunches
    C. No More Homework
    D. Books Tell us How to Teach

37. Why did Petronella try this experiment?
    A. To show more work and more play is fun
    B. To reduce tension for kids
    C. To make parents do more work
    D. To develop an older system of education

38. Which other country tried the same thing in 2018?
    A. Canada
    B. Sweden
    C. Finland
    D. America

39. Finland has stopped teaching _____ at most grades.
    A. behaviour
    B. classes
    C. sports
    D. subjects

40. What is the meaning of the word 'efficacious' in paragraph 6?
    A. Successful
    B. Corrupt
    C. Great
    D. Silly

SPOKEN AND WRITTEN EXPRESSION

FOR QUESTIONS 41 TO 45, CHOOSE THE MOST SUITABLE OPTION TO COMPLETE EACH CONVERSATION.

41. John: I'm at my _____ end and can't find a solution!
    Mother : Don't worry. I'm sure you'll think of something.
    A. road's
    B. school's
    C. wit's
    D. maths'

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42. Alice : This essay is really difficult to start.

Albert : I know but we've _____ get it finished before Monday.

A. got to
B. going to
C. must
D. could

43. Erick : I wish I _____ to the party last night; I'm so tired now.

A. haven't gone
B. wasn't going
C. am not going
D. hadn't gone

44. David : Do you remember playing tennis when we were young?

June : Yes, sure I do, and I don't think I've _____ my touch at all.

A. hidden
B. lost
C. traded
D. evoked

45. Brett : If it hadn't been for the lift we got, we'd never have reached the hotel, _____?

A. won't we
B. hadn't we
C. would we
D. mustn't we

ACHIEVERS SECTION

FOR QUESTIONS 46 AND 47, CHOOSE THE CORRECT WORD/PHRASE TO COMPLETE EACH SENTENCE.

46. The gates will be opened slowly in _____ with the instructions.

A. accordingly
B. accord
C. according
D. accordance

47. I wish he _____ so rude to everyone when we go out.

A. wouldn't be
B. hadn't been
C. won't be
D. didn't be

48. CHOOSE THE CORRECT SYNONYM OF THE GIVEN WORD.

Denounce

A. Tolerate
B. Excoriate
C. Exculpate
D. Repent

49. CHOOSE THE CORRECT ANTONYM OF THE GIVEN WORD.

Altruistic

A. Workable
B. Erroneous
C. Selfish
D. Dreadful

50. CHOOSE THE MOST SUITABLE OPTION TO COMPLETE THE CONVERSATION.

Astrid : My brother has really gone _____ far this time. I have no space left to do my work.

A. badly
B. much
C. a lot
D. too

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